



PROGRAMMING FUNDAMENTALS.

C++ I.

Create a lab report for the following programming challenge:

## Homeowner Expenses Calculator

### Objective

Write a program that calculates and displays the total expenses incurred by a homeowner over a specified range of months using month numbers (e.g., 1 for January, 2 for February). The program should include functions to collect and return the following information:

1. **Range of Months** for which expenses are being calculated (e.g., from month 1 to month 3).
2. **Amount of Mortgage/Rent Payment** (the same amount for each month in the specified range) and the Expected Budgeted Amount.
3. **Home Insurance Costs** (the same amount for each month in the specified range) and the Expected Budgeted Amount.
4. **Utilities Expenses** (electricity, water, gas, internet):
  - a. The program should allow users to input the monthly utility costs, which may vary by month, along with the Expected Budgeted Amount for each utility.
5. **Total Maintenance and Repair Costs** incurred during the entire specified period and the Expected Budgeted Amount.
6. **Total Renovation Costs**, if any, during the entire specified period and the Expected Budgeted Amount.
7. **Total Amount Spent on Meals** Entered monthly for the range of months, with a breakdown for:
  - **Groceries Purchased** (total for each month) and the Expected Budgeted Amount.
  - **Restaurant Purchases** (total for each month) and the Expected Budgeted Amount.

## The program should calculate and display

- The total expenses incurred by the homeowner.
- The total budgeted expenses for the period.
- A list of expenses that were over budget, showing which specific items exceeded their expected budgeted amounts (e.g., Mortgage/Rent Payment, Home Insurance Costs, Utilities, Maintenance and Repair Costs, Renovation Costs, Groceries Purchased, Restaurant Purchases).
- The amount saved by the homeowner if the expenses were under the total budgeted amount.

In your lab report, in an essay style, clearly state in your own words, the different stages of software development for this programming challenge. Analysis, Design, Implementation, Testing.

Create a hand-tracing sheet where the user cases are previously demonstrated and tested. Use the hand-tracing sheet to validate program outputs from the user inputs. In the lab report clearly demonstrate that your program complies with the tested data in the hand tracing sheet.

Place the tracing sheet in your lab report. A screenshot is acceptable.

## Input Validation

- Do not accept negative numbers for any dollar amount.
- Do not accept numbers less than 1 for the range of months.

Additional Requirements:

- Implement modularity as much as possible.
- Design a menu function(s) to improve user-program interaction.
- Implement user input validation to improve software security.
- Provide testing of several cases of the user-interaction.
- Screenshots must contain relevant information about the case you are demonstrating and documenting.
- Your lab report must look professional and complete. Demonstrate your program is robust, reliable, scalable.

You must submit three files.

- A pdf document, the lab report. HomeSpenceCalculator.pdf.
- The .cpp file of your program: HomeSpenceCalculator.cpp
- The excel file with the data of your testing hand-tracing data.

For your reference, there is a lab report template available in the Blackboard system under Course Content.

Any question? Just email me.

LDV.